

Marking Rubric for Lab #2 Assessment (Week 8)

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LAB2 Marking Rubric

		HD	D	C	P	N
Methodology (50%)						
Task 1 Open MPI	Correctness, Performance, and Clarity (15%)	The program is capable to find and print/output <u>a sorted list</u> of prime numbers with Open MPI that are strictly less than an integer n, provided by the user. The code has performed significant optimisation to reduce and eliminate unnecessary computations and bottlenecks, thereby improving parallel speedup. The parallel partitioning scheme and workload distribution tasks are sound and <u>optimal</u>. The overall speedup (including communication, computation, sorting and file writing) is strictly larger than 1 and <u>close to achieving a linear speedup</u> (i.e. speed up close to the number of computing cores available) The coding style follows good industry practices. All the required graphs in the specification are presented without flaws/data omissions. Performance analysis using CAAS or a locally set up cluster is	The program is capable to find and print/output <u>a sorted list</u> of prime numbers with Open MPI which are strictly less than an integer n, provided by the user. The overall speedup (including communication, computation, sorting and file writing) is strictly larger than 1 The code has performed significant optimisation to reduce and eliminate unnecessary computations and bottlenecks to improve parallel speedup. The parallel partitioning scheme and workload distribution tasks are correct. The code is readable and follows good programming practice with sufficient comments. All the required graphs in the specification are presented, with only minor flaws/data omissions. Performance analysis using CAAS or a locally set up cluster is included.	The program is capable to find and print/output <u>a sorted list</u> of prime numbers with Open MPI which are strictly less than an integer n, provided by the user. The overall speedup (including communication, computation, sorting and file writing) is strictly larger than 1 The parallel partitioning scheme and workload distribution tasks are correct, despite may not be optimal. Most the required graphs in the specification are presented and largely correct.	The program is capable to find and print/output prime numbers with Open MPI which are strictly less than an integer n, provided by the user. The parallel partitioning scheme and workload distribution tasks are largely correct.	Task 1 is missing in the presentation, or missing in the submitted presentation material (or) The parallel code contains major flaws affecting the correctness of the program (or) The code completely fails to perform parallelism using Open MPI (or) The code fails to find prime numbers (or) The program print/output values that are not prime numbers (or) The program fails to be properly executed due to major bugs/flaws (including deadlock and system crash)

		included.				
Task 2 Open MPI + OpenMP	Correctness, Performance, and Clarity (15%)	The program is capable to find and print/output <u>a sorted list</u> of prime numbers with Open MPI and Open MP which are strictly less than an integer n, provided by the user. The code has performed significant optimisation to reduce and eliminate unnecessary computations and bottlenecks to improve parallel speedup. The parallel partitioning scheme and workload distribution tasks are sound and <u>optimal</u>. The overall speedup (including communication, computation, sorting and file writing) is strictly larger than 1 and <u>close to achieving a linear speed up</u> (i.e. speed up close to the number of computing cores available). The coding style follows good industry practices. All the required graphs in the specification are presented without flaws/data omissions. Performance analysis using CAAS or a locally set up cluster is included.	The program is capable to find and print/output <u>a sorted list</u> of prime numbers with Open MPI and Open MP which are strictly less than an integer n, provided by the user. The overall speedup (including communication, computation, sorting and file writing) is strictly larger than 1. The code has performed significant optimisation to reduce and eliminate unnecessary computations and bottlenecks to improve parallel speedup. The parallel partitioning scheme and workload distribution tasks are correct. The code is readable and follows good programming practice with sufficient comments. All the required graphs in the specification are presented, with only minor flaws/data omissions. Performance analysis using CAAS or a locally set up cluster is included.	The program is capable to find and print/output <u>a sorted list</u> of prime numbers with Open MPI and Open MP which are strictly less than an integer n, provided by the user. The overall speedup (including communication, computation, sorting and file writing) is strictly larger than 1. The parallel partitioning scheme and workload distribution tasks are correct, despite may not be optimal. Most the required graphs in the specification are presented and largely correct.	The program is capable to find and print/output prime numbers with Open MPI and Open MP which are strictly less than an integer n, provided by the user. The parallel partitioning scheme and workload distribution tasks are largely correct.	Task 2 is missing in the presentation, or missing in the submitted presentation material (or) The parallel code contains major flaws affecting the correctness of the program (or) The code completely fails to perform parallelism using Open MPI + Open MP (or) The code fails to find prime numbers (or) The program print/output values that are not prime numbers (or) The program fails to be properly executed due to major bugs/flaws (including deadlock and system crash)

Task 3 Performance Evaluation	Correctness, Performance, and Clarity (20%)	Theoretical speedup for both Task 1 and Task 2 has been correctly computed with either Amdahl's Law or Gustafson's Law, without any <u>flaws/omissions</u> .	Theoretical speedup for both Task 1 and Task 2 has been correctly computed with either Amdahl's Law or Gustafson's Law.	Theoretical speedup for both Task 1 and Task 2 has been correctly computed with either Amdahl's Law or Gustafson's Law.	Theoretical speedup for both Task 1 and Task 2 has been computed with either Amdahl's Law or Gustafson's Law, despite some errors.	Theoretical speedup is missing (or)
		Time measurements on the serial/parallel part(s) of Task 1 and Task 2 implementations have been correctly performed.	Time measurements on the serial/parallel part(s) of Task 1 and Task 2 implementations have been correctly performed.	Time measurements on the serial/parallel part(s) of Task 1 and Task 2 implementations have been performed, despite minor errors/omissions.	Time measurements on the serial/parallel part(s) of Task 1 and Task 2 implementations have been performed, despite some errors/omissions.	Failed to compute theoretical speedup with either Amdahl's Law or Gustafson's Law (or)
		All the required graphs in the specification are presented without <u>any flaws, errors, or omissions</u> .	All the required graphs in the specification are presented without noticeable errors.	All the required graphs in the specification are presented and largely correct.		Completely failed to perform time measurements on either the serial/parallel part(s) of the Task 1 and Task 2 implementations.
		Comparisons between empirical speedup and theoretical speedup are sound and without any omissions.	Comparisons between empirical speedup and theoretical speedup are largely sound.			
Presentation Quality and Q&A (50%)						
Presentation Quality (Clarity and Organisation) (10%)		The presentation is between 6 to 7 minutes.	The presentation is between 6 to 7 minutes.	The presentation is generally understandable but has major rooms for improvements.	The presentation is somewhat understandable but lacks either organisation or content flow.	The presentation is largely unclear, poorly organised, and hard to follow.
		The presentation is <u>exceptionally</u> clear, well-organised, and easy to follow.	The presentation is clear and organised with minor areas for improvement.			
		The presentation discusses major issues, limitations, and future work <u>well</u> .	The presentation discusses major issues, limitations, and future work.			
		The presentation is concise, precise, and accurate (i.e., materials have been digested, analysed, and chosen carefully without unnecessary information or details).				

	Students are required to extensively use parallel computing terminologies (instead of general computing terms) at this level.				
Q&A (40%)	Capable of understanding and answering all the questions without external tools. All the answers are correct without any flaws. The answers are concise, precise, and accurate (i.e., fully answer the question without unnecessary information or details). Students are required to answer by using parallel computing terminologies (instead of general computing terms) at this level.	Capable of understanding all the questions and answering all the questions without external tools. The answers are all correct without any flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some minor flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some major flaws.	Fails to understand the questions (or) Relies on external tools (e.g., AI, software tools, or online tools) to understand the questions or answer the questions. (or) Answers are completely wrong with major flaws/issues.

Special Consideration & Extensions

- From the special consideration procedure, short extensions are not available for group assessments, in-class assessments (including in-class tests/quizzes and presentations), practical assessments (including laboratories), mid-semester tests, or scheduled final assessments. For these assessment types, students must instead apply for special consideration.
- Unless a special consideration extension is approved, the student group/team will generally receive a 0 mark if they fail to attend the in-class session, fail to demonstrate the work, or fail to submit the required files to Moodle.
- For students with special considerations approved by the special consideration team, an alternative form of assessment must be submitted no later than the newly approved deadline. Students are reminded to read and follow the special consideration and alternate assessment sections in the assessment specification.
- If the extended due date falls after Week 14 and you plan to submit on/after Week 14, students are advised to contact the Chief Examiner and/or the Unit Coordinator, as grades may need to be withheld to allow extra time for marking.
- Students who have received special considerations but failed to submit the required files to Moodle by the newly approved deadline for their alternative assessment will be treated as having made no submission (or as having submitted late, if the file eventually arrives after the newly approved due date). For alternative assessments, late penalties will still apply for late submissions.